

Informed Consent Form for Participation in Experiments on the Influence of Street Network Topological Regularity on the Spatial Cognition of Individuals with Multiple Sclerosis (MS)

Dear Participant,

You are cordially invited to participate in the aforementioned research. The relevant details regarding this study are provided in this document, and your participation is entirely voluntary.

You are under no obligation to make an immediate decision; you may address any questions to the research team and consult with any individual of your choosing. Prior to signing this informed consent form, please ensure that you have fully comprehended all the information provided and that all your inquiries have been satisfactorily answered.

Hamed Ahmadi

1-I understand that the objectives of this study are as follows:

To investigate the influence of street network topological regularity on the wayfinding performance of individuals with Multiple Sclerosis (MS). Specifically, the study aims to evaluate whether the level of topological regularity within the street network significantly impacts the wayfinding abilities of individuals with MS.

2-I understand that my participation in this research is entirely voluntary and that I am under no obligation to participate. I have been assured that declining to participate will not result in the deprivation of my standard diagnostic or therapeutic care, nor will it adversely affect my professional relationship with the medical center or my attending physician.

3-I am aware that even after consenting to participate, I may withdraw from the study at any time by notifying the investigator. Such withdrawal will not jeopardize my access to routine medical services.

4-My participation in this research will be conducted as follows:

Using a laptop and its peripherals (mouse and keyboard), which will be provided by the investigator, I will navigate from a designated starting point to three specified destinations in a predetermined sequence. This experiment will take place within a room. Prior to the task, I will be given an opportunity to familiarize myself with both the simulated environment and the equipment. Additionally, I will complete a questionnaire regarding personal characteristics (including age, gender, educational level and field, familiarity with gaming environments, severity of the disease, type of MS, duration of the illness, medication use and type, etc.) and the characteristics of the task performed (including perceived difficulty and environmental complexity). The experiment will be conducted in a single session, lasting approximately 70 minutes. There will be no medical interventions, paraclinical procedures, or biological sampling.

5-The potential benefits of my participation in this study are as follows:

This research receives no financial funding from any organization or entity; consequently, no monetary compensation will be provided to participants. However, in alignment with the research objectives, the findings will assist urban managers, planners, and designers in improving the design and planning of urban environments. By integrating the specific needs of individuals with MS into the design of street networks, the accessibility and wayfinding experience for this demographic can be significantly enhanced.

6-The potential risks and adverse effects of participating in this study are as follows:

As the experiment is conducted in a controlled indoor setting rather than a real-world urban environment, utilizing non-invasive tools (laptop, mouse, and keyboard), there are no anticipated risks or adverse effects for the participants, neither during nor after the study.

7-In the event that I choose not to participate, the standard course of care will be provided. The implications are as follows:

Given that MS can affect memory and cognitive functions, and based on existing research, affected individuals may encounter significant challenges in wayfinding. Urban planning and design that accounts for these challenges can mitigate or resolve such issues. Without the participation of individuals with MS in this study, it will remain difficult to accurately identify these challenges and, subsequently, to address them within urban planning and design frameworks.

8-I understand that the investigators will maintain all my personal information as strictly confidential and are permitted to publish only the aggregate and group results of this research without disclosing my name or identity.

9-I am aware that the Research Ethics Committee may access my information for the purpose of monitoring the protection of my rights.

10-I understand that I will not be responsible for any costs associated with the research interventions. As previously stated, this experiment involves no medical interventions and incurs no financial cost for the participants.

11-Mr. Hamed Ahmadi has been introduced to me as the point of contact for any queries or concerns regarding my participation in this research. His contact details have been provided as follows:

Address: No. 8 (Pardis Building), 2nd Floor, 1st Alley, Niloufar Complex, Shahid Barghi St., Khatib St., Tabriz

Landline: 04134475622 | Mobile: 09144939124

12-I understand that should any physical or psychological complications arise during or after the research as a direct result of my participation, the treatment of such complications,

the associated costs, and any relevant compensation shall be the responsibility of the investigator.

13-I am aware that should I have any grievances or objections regarding the researchers or the conduct of the study, I may contact the Research Ethics Committee of the Tehran University of Medical Sciences. I can express my concerns either verbally or in writing via the following address: Room 501, 5th Floor, Central Headquarters Building of Tehran University of Medical Sciences, Intersection of Keshavarz Blvd. and Qods St., Tehran.

14-This information and informed consent form has been prepared in duplicate; following the signature, one copy will be provided to me and the other will be retained by the investigator.

I have read and understood the aforementioned points, and based on this, I hereby provide my informed consent to participate in this research.

Participant's Signature

I, Hamed Ahmadi, acknowledge my obligations as the investigator as set forth in the aforementioned terms and pledge to ensure the protection of the participants' rights and safety throughout this research.

Investigator's Stamp and Signature



University of Tehran



Faculty of Psychology and
Education

Research Ethics Committees Certificate

Approval ID:	IR.UT.PSYEDU.REC.1404.015	Approval Date:	2025-04-28
Evaluated by:	Research Ethics Committees of Faculty of Psychology and Education		
Status:	Approved		
Approval Statement:	The project was found to be in accordance to the ethical principles and the national norms and standards for conducting Medical Research in Iran. Notice: 1. Although the proposal has been approved by the Biomedical Research Ethics Committee, meeting the professional and legal requirements is the sole responsibility of the PI and other project collaborators. 2. This certificate is reliant on the proposal/documents received by this committee on 2025-04-28. The committee must be notified by the PI as soon as the proposal/documents are modified.		
Thesis Title:	Geographical parameter analysis in spatial cognition (wayfinding of people with disabilities)		
Supervisor:	Name: Meysam Argany Email: argany@ut.ac.ir		
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